Eduardo Mucelli — Senior Software Engineer

PhD in Computer Science

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I build backend systems and AI infrastructure, drawing on research in large-scale data analysis and modeling. I'm comfortable working closely with research teams to evaluate ideas and bring them into production.

Education

École Polytechnique — **France**
Computer Science, PhD — *October 2011 — May 2015*
Research on human mobility and large-scale network traffic using Python/R. Produced 5 conference papers, 2 journal papers, and a data traffic simulator. Thesis: *From Your Routine to Better Network Services*.

Federal University of Minas Gerais — **Brazil**
Computer Science, Master's degree — *March 2009 — July 2011*
Thesis: *Centrality-based Routing for Wireless Sensor Networks*.

Pontifical Catholic University of Minas Gerais — **Brazil**
Computer Science, Bachelor's degree — *August 2004 — August 2008*

Professional experience

Synthesia — **Remote**
Senior Software Engineer — *January 2023 – Present*
Improved the robustness, scalability, and observability of Synthesia's AI video generation platform, leading work on rate limiting, task scheduling, diagnostics, rendering/preprocessing workers, GPU rendering experiments, and dubbing infrastructure. Led scaling and cost-efficiency experiments that reduced compute usage by ∼56% with minimal user impact, identifying an estimated ∼$800K/year in infrastructure savings.

Canonical — **Remote**
Software Engineer / Software Engineer II — *June 2020 – November 2022*
Implemented kernel-module signature verification and failure reporting in Canonical Livepatch's Go client, with corresponding server-side patch-state support. Developed Go services for Ubuntu Advantage contract provisioning and renewals, optimizing PostgreSQL queries and API usage. Refactored Python ETL tooling with typed interfaces, retries, and monitoring; maintained Kubernetes/Argo workflows and PostgreSQL snapshots. Introduced Gherkin/Godog acceptance tests in CI.

BlaBlaCar — **France**
Software Engineer — *July 2016 – May 2020*
Developed Java services for BlaBlaCar's Trip Search team, covering search, routing, and matching. Built the meeting-point service with PostgreSQL/PostGIS spatial search, machine-learning ranking, Redis, and R-tree caching. Created a geospatial crawler and ingestion tools for over one million collected place records across 14 countries. Benchmarked JVM inference engines and built a reusable XGBoost prediction library for Java services. Improved geospatial utilities, search indexing, and routing integrations.

Orange — **France**
Postdoctoral Researcher — *June 2015 – June 2016*
Developed machine learning methods to predict video streaming quality of experience from mobile network performance metrics.

Dito Internet — **Brazil**
Software Engineer — *July 2011 — September 2011*
Developed TIM Beta for Telecom Italia Mobile as part of a Ruby on Rails team.

Telecom Italia Future Centre — **Italy**
Research Intern — *November 2010 — May 2011*
Developed Future of Enterprises in Python to improve real-time collaboration among employees.

Task Internet — **Brazil**
Software Engineer — *February 2008 — February 2009*
Led development of a Ruby on Rails social network.

Selected projects

Docking

A lightweight, feature-rich dock for Linux and BSD written in Python with GTK 3 and Cairo.

CEP Aberto

Collaborative platform to open Brazilian Postal Code (CEP) data, with information for about 1 million CEPs. Provides both a developer API and an end-user collaborative platform to improve data quality, with 50k+ users.

Rosto.io

Face recognition SaaS providing detection and analysis features such as age, gender, emotion, ethnicity, landmarks, glasses, smile, and hair segmentation. Built on end-to-end machine learning models and monitored with production-grade observability.

Eazy.bike, Android app, iOS app

Bike-sharing route planner in 443+ cities across 45 countries. Uses real-time bike availability and station proximity to suggest pick-up/drop-off points, with mobile apps automatically redirecting when the destination station fills up.

Proconfie

Application helping users choose companies based on customer complaints.

Benchmark XGBoost Java

Comprehensive benchmark of XGBoost libraries for Java.

Synthetic Network Traffic Generator

Synthetic traffic generator created from real data from more than 6.8 million subscribers in Mexico City.

Skills

Language: Python, Golang, Java, Ruby, R

Framework: Spring Boot, Flask, FastAPI, Ruby on Rails

Machine learning: Keras, PyTorch, Caret, XGBoost

OS: Linux

Infrastructure: Docker, Nginx, Prometheus, Grafana

DB: MongoDB, PostgreSQL (PostGIS), Redis, Elasticsearch

Selected publications and technical writing

- Fast Machine Learning Predictions, BlaBlaCar Tech Blog, October 2017.
- Machine Learning for Predicting QoE of Video Streaming in Mobile Networks, IEEE ICC Communication QoS, reliability and modeling symposium, May 2017.
- Mobile Data Traffic Modeling: Revealing Temporal Facets, Elsevier Computer Networks, November 2016.
- On the regularity of human mobility, Elsevier Pervasive and Mobile Computing, May 2016.

Languages

Portuguese: Native speaker

French: Fluent

English: Fluent

Italian: Advanced

Awards

Honorable mention
Brazilian Ministry of Justice — *May 2013*
Lead developer of Proconfie, recognized in the Open Data Applications Contest.

Silver Medal
Pontifical Catholic University of Minas Gerais — *July 2008*
Ranked second overall in the bachelor's degree program.

Honorable mention
Pontifical Catholic University of Minas Gerais — *May 2007*
For the work *"Game Theory for Decision Making in Wireless Sensor Networks"*, presented at the 15th Science Seminar.